

Cognitive Models of Semantic Representation

Lecture 11: How Humans Represent Meaning

PSYC 51.07: Models of Language and Communication - Week 4

Winter 2026

Today's Lecture

1.  **Human vs. Computational Semantics**
2.  **Cognitive Theories of Meaning**
3.  **Embodied & Grounded Cognition**
4.  **Semantic Similarity: What Does It Mean?**
5.  **Empirical Evidence from Cognitive Science**
6.  **Bridging the Gap: Models Meet Minds**

Goal: Understand what computational models are really learning

What does "meaning" actually mean?

Computational View:

- Vectors in high-dimensional space
- Learned from text co-occurrence
- Distributional patterns
- Statistical relationships
- "You shall know a word by the company it keeps"

But is this how humans represent meaning?

Human View:

- Sensory experiences
- Emotional associations
- Physical grounding
- Social context
- Multimodal integration
- Embodied understanding

Can text-only models capture this?

This lecture explores the relationship between computational and cognitive semantics

How Do Humans Represent Meaning?

Example: The word "coffee"

What comes to YOUR mind?

- **Visual:** brown liquid, mug, steam, beans
- **Olfactory:** aroma, roasted smell
- **Gustatory:** bitter taste, smooth texture
- **Tactile:** hot, warm cup, liquid
- **Auditory:** brewing sounds, pouring
- **Motor:** lifting cup, drinking motion
- **Contextual:** morning routine, work, café
- **Emotional:** comfort, alertness, pleasure
- **Social:** conversations, meetings
- **Cultural:** Starbucks, espresso, traditions

Word2Vec representation:

coffee = [0.23, -0.45, 0.67, 0.12, -0.89, ...]

(300 numbers learned from text)

What's missing?

- No sensory grounding
- No embodied experience
- No emotional content
- No perceptual features
- No action affordances
- Purely linguistic

The Symbol Grounding Problem

Harnad (1990): Can symbols have intrinsic meaning?

The Chinese Room Argument (Searle, 1980)

- Person in room, doesn't understand Chinese
- Receives Chinese symbols, has rule book
- Follows rules to produce Chinese responses
- From outside: appears to understand Chinese
- But has no understanding of meaning!

Analogy: Language models manipulate symbols without grounding

Symbol Systems:

- Symbols refer to other symbols
- Purely syntactic manipulation
- No connection to world
- "Ungrounded" semantics

Grounded Systems:

- Symbols connected to perception
- Linked to action
- Embodied in physical world
- "Intrinsic" semantics

References: Searle (1980). "Minds, Brains, and Programs"; Harnad (1990). "The Symbol Grounding Problem"

Embodied Cognition Theory

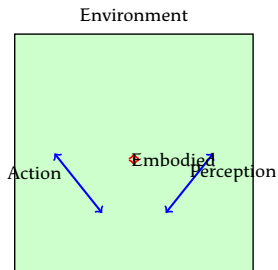
Meaning arises from bodily experience and sensorimotor interaction

Key Principles:

1. **Embodiment:** Cognition shaped by body
2. **Situatedness:** Meaning context-dependent
3. **Enactivism:** Knowing through doing
4. **Grounding:** Concepts tied to perception/action

Evidence:

- Reading "kick": activates motor cortex
- Understanding "grasp": simulates action
- Processing emotions: activates body maps
- Conceptual metaphors based on physical experience



Implications for AI:

- Text-only models miss embodiment
- Need multimodal grounding
- Interaction with environment
- Sensorimotor foundations

Reference: Barsalou (2008). "Grounded Cognition"

The Distributional Hypothesis Revisited

"You shall know a word by the company it keeps"

— J.R. Firth (1957)

Strong version: Word meaning IS distributional patterns

Weak version: Distributional patterns REFLECT meaning

Supports:

- Works remarkably well in practice
- Captures semantic similarity
- Enables analogical reasoning
- Scales to huge vocabularies
- Unsupervised learning
- Aligned with usage-based linguistics

Limitations:

- Correlation $\neq$ causation
- Lacks perceptual grounding
- No embodied understanding
- Missing common sense
- Can't handle novel situations
- Reflects training data biases

Key Question

Is distributional semantics sufficient for meaning, or just a useful approximation?

Reference: Boleda (2020). *"Distributional Semantics and Linguistic Theory"*

What Does "Semantic Similarity" Really Mean?

Different types of similarity:

1. Taxonomic:

- dog $\sim$ cat (both animals)
- red $\sim$ blue (both colors)
- "Is-a" relationships

2. Thematic:

- dog $\sim$ leash (co-occur in events)
- coffee $\sim$ cup (functional relation)
- "Goes-with" relationships

3. Synonymy:

- big $\sim$ large (same meaning)
- happy $\sim$ joyful

4. Functional:

- knife $\sim$ scissors (similar function)
- chair $\sim$ stool

What do models capture?

- Word2Vec: Mostly thematic + taxonomic
- BERT: Better at taxonomic
- Varies by context window size
- Training data matters!

Human judgments:

- SimLex-999: True similarity
- WordSim-353: Relatedness (broader)
- Models better at relatedness than similarity

Example

Similar: car $\sim$ automobile (0.9)

Related: car $\sim$ road (0.7)

Models often conflate these!

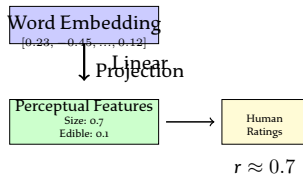
Reference: Hill et al. (2015). "SimLex-999: Evaluating Semantic Models with Genuine Similarity Estimation"

Semantic Projection: Recovering Human Knowledge

Grand et al. (2022): Can we extract human-like features from embeddings?

The Experiment:

1. Collect human ratings on perceptual features
 - Is it edible?
 - Is it heavy?
 - Is it alive?
 - Can you hold it?
2. Train linear projections from word embeddings
3. Test if embeddings predict human ratings



Key Findings:

- Embeddings encode surprisingly rich knowledge
- Can predict perceptual features
- Even vision and motor properties!
- Better with larger models

Reference: Grand et al. (2022). "Semantic projection recovers rich human knowledge of multiple object features from word embeddings"

Interpretation:

- Text co-occurrence captures real-world properties
- Indirect grounding through language
- But still not true perceptual grounding

Multimodal Models: Bridging the Gap

Combining language with perception

Vision-Language Models:

- CLIP (OpenAI)
- ALIGN (Google)
- Flamingo (DeepMind)
- GPT-4V (OpenAI)

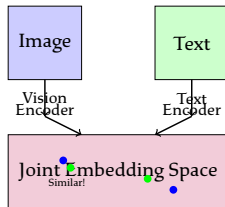
Key Idea:

- Learn joint embedding space
- Text and images map to same space
- Enables cross-modal understanding
- Grounds language in vision

Training:

- Image-caption pairs
- Contrastive learning
- Matching images to descriptions
- Large-scale (400M+ pairs)

Reference: Radford et al. (2021). "Learning Transferable Visual Models From Natural Language Supervision" (CLIP)



Benefits:

- Visual grounding
- Zero-shot classification
- Better generalization
- More "human-like"

Conceptual Spaces Theory

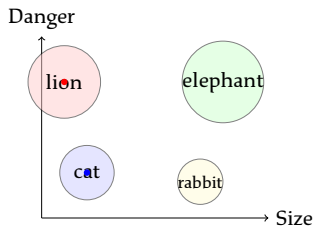
Gärdenfors (2000): Meaning as geometry

Key Ideas:

- Concepts represented in quality dimensions
- Dimensions: color, size, temperature, etc.
- Each dimension has a metric
- Concepts are regions in space
- Similarity = geometric proximity

Example: Colors

- Hue, saturation, brightness
- Natural categories (red, blue, green)
- Fuzzy boundaries
- Prototypes at centers



Relation to Embeddings:

- Similar geometric structure
- But learned dimensions
- Not interpretable qualities
- No grounding in perception

Reference: Gärdenfors (2000). "Conceptual Spaces: The Geometry of Thought"

How do linguists think about word meaning?

1. Feature-Based:

- Words = bundles of features
- bachelor = [+human, +male, +adult, -married]
- Compositional
- Logical

2. Prototype Theory:

- Categories have best examples
- Robin is a prototypical bird
- Penguin is peripheral
- Graded membership

3. Frame Semantics:

- Words evoke conceptual frames
- "Buy" activates commerce frame
- Buyer, seller, goods, money
- FrameNet resource

4. Construction Grammar:

- Meaning from form-function pairings
- Idioms, patterns
- Usage-based

5. Word Sense Disambiguation:

- Words have multiple senses
- WordNet: hierarchical ontology
- Bank₁ (financial), Bank₂ (riverside)

Question

Which theories align with distributional models?

Answer: Mostly usage-based views
(Construction Grammar, Prototype Theory)

Struggle with: Feature analysis, Frame semantics

References: Fillmore (1982). "Frame Semantics"; Rosch (1975). "Cognitive Representations of Semantic Categories"

Empirical Evidence from Neuroscience

What does the brain tell us about semantic representation?

fMRI Studies:

- Can predict brain activity from word embeddings
- Semantic information distributed across cortex
- Different regions for different features
- Temporal lobe: objects
- Motor cortex: actions
- Visual cortex: visual features

Findings:

- Word2Vec correlates with neural patterns
- But: imperfect match
- Human brain uses multimodal integration
- Language areas + sensory areas

References: Mitchell et al. (2008). "Predicting Human Brain Activity Associated with the Meanings of Nouns";
Huth et al. (2016). "Natural speech reveals the semantic maps that tile human cerebral cortex"

Brain vs. Model Representations:

Feature	Brain	Model
Multimodal	☐	☐
Distributed	☐	☐
Hierarchical	☐	☐
Context-sensitive	☐	☐ (BERT)
Grounded	☐	☐
Fast	☐	☐

Key Insight:

- Models capture some aspects
- But miss crucial grounding
- Convergent representation learning?
- Or fundamentally different?

The "Stochastic Parrots" Debate

Bender et al. (2021): On the Dangers of Stochastic Parrots

The Argument:

- LLMs learn form, not meaning
- "Stochastic parrots" - repeating patterns
- No understanding of world
- No communicative intent
- Risk: Mistaking fluency for understanding

Evidence:

- Fail on simple reasoning
- Sensitive to phrasing
- Hallucinate facts
- No common sense
- Brittle to adversarial inputs

Counter-Arguments:

- Emergent capabilities at scale
- Performance on complex tasks
- Transfer to new domains
- Maybe understanding = prediction?
- Pragmatic success criterion

The Core Question

Can meaning arise from form alone?

Or do we need grounding in:

- Perception?
- Action?
- Social interaction?
- Physical embodiment?

References: Bender & Koller (2020). "Climbing towards NLU"; Bender et al. (2021). "On the Dangers of Stochastic Parrots"

Common Sense Reasoning 🤖

What humans know but models don't

Examples:

Physical

- Water is wet
- Objects fall down
- You can't be in two places at once
- Heavy things are hard to lift

Social

- People have feelings
- Insults are hurtful
- Promises should be kept
- Eye contact shows attention

Why Models Struggle:

- Not explicit in text
- Assumed background knowledge
- Requires world experience
- Needs causal reasoning
- Embodied understanding

Benchmark Datasets:

- Winograd Schema Challenge
- PIQA (Physical Interaction QA)
- SocialIQA
- CommonsenseQA

Progress:

Large models do better

Compositionality: Phrases and Sentences

How do we combine word meanings?

The Problem:

- "hot" + "dog" $\neq$ "hot dog"
- "red" + "herring" $\neq$ "red herring"
- Idioms, metaphors, collocations
- Non-compositional meaning

Classical Approaches:

- Vector addition: $\vec{hot} + \vec{dog}$
- Element-wise multiplication
- Tensor products
- Fails for non-compositional phrases!

Neural Approaches:

- RNNs, LSTMs
- Transformers (BERT)

Compositionality Types:

1. Semantic Composition:

- "big red ball" = $\text{big} \cap \text{red} \cap \text{ball}$
- Intersective

2. Non-intersective:

- "fake gun" $\neq \text{fake} \cap \text{gun}$
- Adjective changes interpretation

3. Metaphorical:

- "Time is money"
- Cross-domain mapping

Current Status

Do large language models "understand" language?

Arguments FOR:

- Solve complex tasks
- Generalize to new domains
- Show emergent capabilities
- Capture linguistic structure
- Pragmatic criterion: if it works...
- Maybe understanding = prediction
- Human understanding also imperfect

"The question is not whether machines think, but whether they behave intelligently" - Turing

Arguments AGAINST:

- No grounding in reality
- No embodied experience
- No intentionality
- Brittle, exploit shortcuts
- Hallucinate confidently
- No causal reasoning
- Missing common sense
- Form without meaning

"Understanding requires grounding in perception and action" - Embodied cognition

Perhaps the wrong question?

Instead of "Do they understand?", ask:

- What do they represent?
- How does it differ from humans?
- What are the limits?
- How can we improve?

Bridging computational and cognitive semantics

1. **Multimodal Learning:**

- Vision + language (CLIP, Flamingo)
- Audio, tactile, proprioception
- Grounding in multiple modalities

2. **Embodied AI:**

- Robots learning through interaction
- Simulation environments
- Sensorimotor grounding

3. **Neurosymbolic AI:**

- Combining neural and symbolic
- Logical reasoning + learning
- Structured knowledge graphs

4. **Cognitively-Inspired Architectures:**

- Memory systems
- Attention mechanisms
- Episodic learning
- Meta-learning

5. **Social Grounding:**

- Learning through dialogue
- Cultural context
- Pragmatic understanding

Practical Implications

What this means for NLP practitioners

1. Know the Limits:

- Models excel at pattern matching
- Struggle with true reasoning
- Need task-specific evaluation
- Don't assume understanding

2. Choose Appropriate Tasks:

- Good: classification, retrieval, similarity
- Moderate: summarization, translation
- Challenging: reasoning, planning, common sense

3. Augment with Structure:

- Knowledge graphs
- Rules and constraints
- Domain expertise
- Human-in-the-loop

4. Evaluate Carefully:

- Beyond accuracy metrics
- Test edge cases
- Adversarial robustness
- Bias and fairness
- Interpretability

5. Stay Informed:

- Rapidly evolving field
- New capabilities emerging
- Ethical considerations
- Interdisciplinary insights

What we learned today:

1. **Symbol Grounding:** Computational models lack perceptual grounding
2. **Embodied Cognition:** Human meaning tied to bodily experience
3. **Distributional Semantics:** Powerful but incomplete theory
4. **Semantic Similarity:** Multiple types, models capture some
5. **Empirical Evidence:** Models align with neural patterns but miss multimodality
6. **Current Debates:**
 - Stochastic parrots vs. emergent understanding
 - Form vs. meaning
 - Prediction vs. comprehension
7. **Future:** Multimodal, embodied, neurosymbolic approaches

**The gap between computation and cognition remains,
but we're making progress! **

Key References

Foundational Papers:

- Searle, J. (1980). "Minds, Brains, and Programs"
- Harnad, S. (1990). "The Symbol Grounding Problem"
- Barsalou, L. (2008). "Grounded Cognition"
- Gärdenfors, P. (2000). "Conceptual Spaces: The Geometry of Thought"

Distributional Semantics:

- Firth, J.R. (1957). "A Synopsis of Linguistic Theory"
- Boleda, G. (2020). "Distributional Semantics and Linguistic Theory"
- Hill et al. (2015). "SimLex-999"
- Grand et al. (2022). "Semantic Projection Recovers Rich Human Knowledge"

Critical Perspectives:

- Bender & Koller (2020). "Climbing towards NLU: On Meaning, Form, and Understanding"
- Bender et al. (2021). "On the Dangers of Stochastic Parrots"

Neuroscience:

- Mitchell et al. (2008). "Predicting Human Brain Activity"
- Huth et al. (2016). "Natural Speech Reveals Semantic Maps"

Multimodal:

- Radford et al. (2021). "Learning Transferable Visual Models" (CLIP)

**As we build ever-larger language models,
let's not forget to ask:**

What are they really learning?

How does it compare to human understanding?

What's missing?

How can we do better?